

TPM and BitLocker

Hardware data protection and encryption technologies

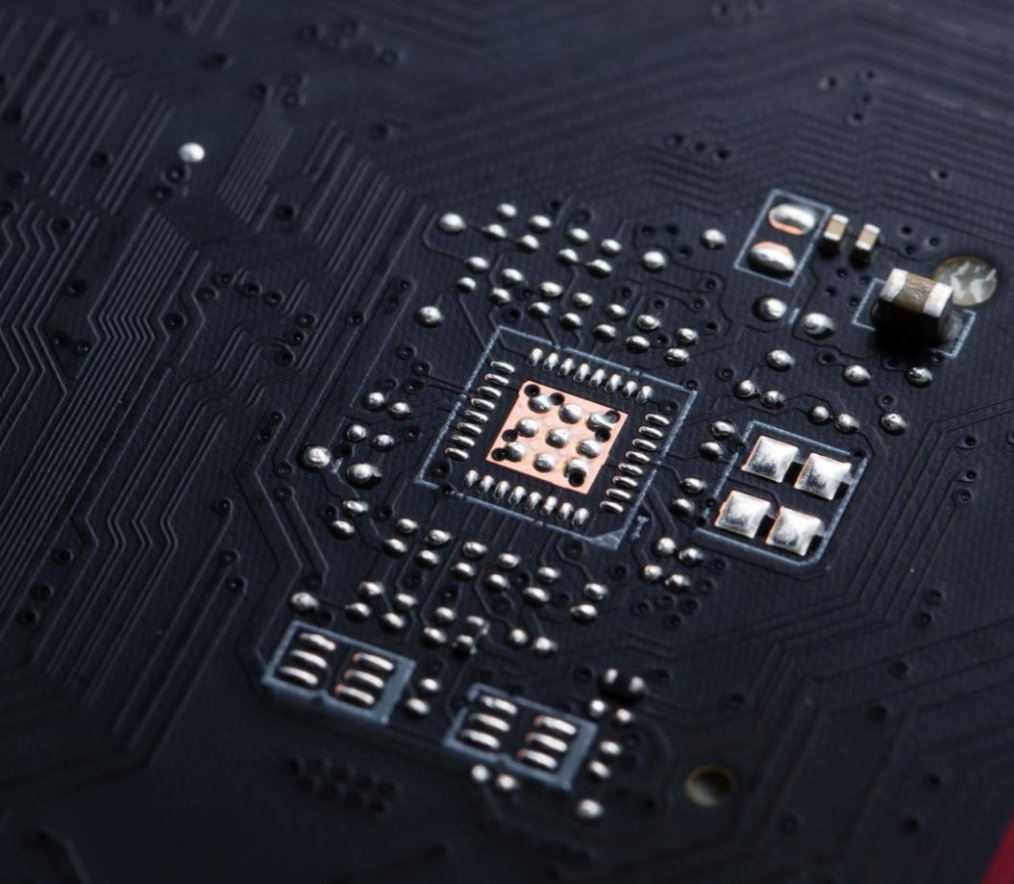
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General Overview



Trusted Platform Module

Hardware-Based Security Isolation

TPM provides hardware-based security located on the motherboard or integrated on the CPU.

Secure Cryptographic Key Management

TPM securely generates, stores, and manages encryption keys, passwords, and digital certificates.

Mitigation of Attacks

TPM restricts brute-force attempts and defends against firmware tampering to enhance system integrity.

Foundation for Trustworthy Systems

TPM ensures device identity and system integrity from boot to operation, supporting compliance and encryption.



BitLocker Encryption and TPM Integration

System Integrity Verification

TPM verifies system integrity by checking firmware and boot components before releasing encryption keys.

Protection Against Preboot Attacks

BitLocker and TPM prevent preboot malware and boot-sector tampering by ensuring secure startup authentication.

Safeguarding Data on Stolen Devices

BitLocker ensures encrypted drives remain inaccessible without TPM validation even if physically stolen or removed.

Hardware-Isolated Key Protection

TPM securely stores encryption keys, isolating them from OS memory and mitigating brute-force attacks.

Organization Implementation



Organizational Use Cases for TPM and BitLocker

Protecting Sensitive Data

TPM and BitLocker protect high-risk data on endpoints, securing confidential information even when devices are offline or exposed.

Compliance with Regulations

They help organizations meet industry standards like GDPR and HIPAA by simplifying encryption requirements and preventing penalties.

Hardware Readiness

Successful TPM and BitLocker deployment requires devices with TPM chips or integrated support, necessitating hardware upgrades if needed.

Strengthening Cybersecurity

Integrating TPM and BitLocker enhances zero-trust security and protects remote work environments from unauthorized data access.